

IR Analysis

1. Precision at 10 (P@10): Best system was System 3 (Mean P@10 = 0.410). Second best – System 5 (Mean P@10 = 0.410). Significance: $p \approx 1.000 \rightarrow$ not statistically significant. There is therefore no evidence that either system is better in terms of P@10.

2. Recall at 50 (R@50): Best system was System 2 (Mean R@50 = 0.867). Second best – System 1 (Mean R@50 = 0.834). Significance: $p \approx 0.703 \rightarrow$ not statistically significant. Although System 2 has a slightly higher mean recall, the difference (0.867 vs 0.834) is small relative to the per-query variability. With only 10 query scores per system, the two-sample t-test does not detect a reliable difference at the 0.05 level.

3. r-precision: Best system was System 3 (Mean r-precision = 0.448). Second best – System 6 (Mean r-precision = 0.448). Significance: $p \approx 1.000 \rightarrow$ not statistically significant.

4. Average Precision (AP): Best system was System 3 (Mean AP = 0.451). Second best – System 6 (Mean AP = 0.445). Significance: $p \approx 0.967 \rightarrow$ not statistically significant. Given the small sample size (10 queries) and variation across queries, this difference is not statistically reliable.

5. nDCG@10: Best system was System 3 (Mean nDCG@10 = 0.420). Second best – System 6 (Mean nDCG@10 = 0.400). Significance: $p \approx 0.882 \rightarrow$ not statistically significant. The t-test indicates this could easily be due to random variation over queries.

6. nDCG@20: Best system was System 3 (Mean nDCG@20 = 0.511). Second best – System 6 (Mean nDCG@20 = 0.491). Significance: $p \approx 0.869 \rightarrow$ not statistically significant. Same as before.

Overall, System 3 performs best on P@10, r-precision, AP, nDCG@10, and nDCG@20 (sometimes tied), with System 2 performing best on R@50. For none of the metrics does the best system differ significantly from the second-best system at $p < 0.05$. Hence, we cannot claim that any system is definitively superior to its closest competitor in a statistically significant sense given this data.

Token Analysis

Mutual Information and chi square diverge because they capture different kinds of statistical distinctiveness. Mutual Information identifies how informative a word is for distinguishing one corpus from another, even when the word appears only infrequently. It rewards distributional surprise, elevating terms that occur disproportionately in a single corpus. Chi square, by contrast, heavily favours high-frequency items: when a word appears many times in one corpus and only rarely elsewhere, its chi square value becomes extremely large. As a result, MI highlights the most informative discriminators, while chi square highlights the most frequently repeated discriminators.

The resulting rankings reveal clear theological and cultural signatures across the three corpora. The Quran is characterised by references to Allah, Muhammad, believers and unbelievers, and eschatological themes. The Old Testament foregrounds covenant theology, national identity, and the narrative of Israel, reflected in terms such as lord, Israel, Moses and Egypt. The New Testament emphasises Christological themes and apostolic figures, including Jesus, Christ, the disciples, Paul and Peter. Together, these patterns show that MI brings out more specific conceptual markers, while chi square elevates high-frequency anchor terms, cleanly separating the corpora along expected theological and historical lines.

Below are tables containing top 10 highest scoring words for each method for each corpus.

Quran:

Rank	MI Token	MI Score	χ^2 Token	χ^2 Score
1	god	0.027	allah	5895.704
2	muhammad	0.021	muhammad	2934.042
3	certainly	0.018	punishment	2553.341
4	believers	0.018	messenger	2198.680
5	torment	0.015	believers	1914.816
6	shall	0.012	mercy	1863.382
7	israel	0.011	righteous	1674.458
8	quran	0.010	shall	1621.244
9	king	0.010	truth	1559.764
10	unbelievers	0.010	guidance	1552.436

Old Testament:

Rank	MI Token	MI Score	χ^2 Token	χ^2 Score
1	jesus	0.027	israel	3181.775
2	shall	0.027	lord	2324.875
3	israel	0.025	egypt	2005.134
4	lord	0.023	moses	1751.449
5	god	0.020	king	1284.659
6	land	0.017	god	1105.017
7	moses	0.015	christ	750.862
8	egypt	0.015	nt	664.625
9	earth	0.014	land	636.108
10	man	0.014	children	617.338

New Testament:

Rank	MI Token	MI Score	χ^2 Token	χ^2 Score
1	jesus	0.041	jesus	2919.947
2	christ	0.034	christ	1795.875
3	disciples	0.026	nt	1589.617
4	peter	0.022	shall	1053.193

5	paul	0.020	lord	968.115
6	church	0.020	things	796.667
7	kingdom	0.019	ot	733.932
8	mary	0.018	disciples	728.643
9	heaven	0.017	peter	526.316
10	spirit	0.016	paul	522.760

Topic Analysis

The topics were chosen using the average document-topic probability per corpus. According to our results in topics_by_corpus.txt, these were:

- NT: topic 2 (average topic score 0.134); my interpretation: **“Jesus dialogue”**.
- OT: topic 9 (average topic score 0.122); my interpretation: **“Lord and Israel”**.
- Quran: topic 1 (average topic score 0.340); my interpretation: **“Belief and warning”**.

Below, for each corpus, the table shows the ten highest-probability tokens from the topic that has the highest average document–topic score within that corpus.

	NT – Topic 2		OT – Topic 9		Quaran – Topic 1	
Rank	Token	Prob.	Token	Prob.	Token	Prob.
1	said	0.0132	lord	0.0141	god	0.0159
2	jesus	0.0128	god	0.0136	people	0.0148
3	answered	0.0106	shall	0.0119	say	0.0133
4	man	0.0091	says	0.0110	lord	0.0129
5	know	0.0084	ot	0.0104	muhammad	0.0122
6	let	0.0080	land	0.0100	truth	0.0116
7	did	0.0078	come	0.0097	certainly	0.0113
8	come	0.0074	people	0.0095	day	0.0109
9	saw	0.0069	israel	0.0093	believers	0.0105
10	say	0.0068	nations	0.0087	believe	0.0103

What LDA model tells us about the corpus:

The LDA model reinforces the thematic separation already suggested by the MI and χ^2 analyses, but it also shows how each corpus blends shared religious vocabulary with corpus-specific emphases. The dominant topics clearly align with the central themes of each text – a Jesus-centred dialogue in the New Testament, covenantal Lord-Israel relations in the Old Testament, and belief and eschatological warning in the Quran – while still revealing overlaps through common terms such as “god”, “lord”, or “people”. By examining topic distributions, it becomes clear that some themes are shared more strongly between certain corpora – such as narrative and prophetic vocabulary linking the OT and Quran, or references to figures like Jesus and Mary linking the Quran and NT

– while others are broadly moral or eschatological across all three. Unlike MI and χ^2 , which highlight words that differentiate the corpora, LDA captures this mixture structure: it uncovers broader latent themes, shows how they vary in strength across the corpora, and exposes both shared foundations and distinct theological patterns that co-exist within the texts.

Classification

Misclassified Examples:

1. **Tweet:** *“Messi has scored against every South American team, he is just the 4th player to achieve that. <http://t.co/c73n9C3Fhn>”*

Gold label: positive; **Predicted:** neutral;

Hypothesis: This tweet expresses admiration and achievement, which is a positive sentiment. However, because the text contains no explicitly emotional adjectives (“amazing”, “great”, etc.), a simple BOW model may treat this as purely factual sports reporting. With no strong sentiment cues, the model defaults to the majority class (neutral). The external link also adds noise for BOW features.

2. **Tweet:** *“just checked for events happening at Gila River Arena in Sept... SO excited to see that Sam Smith is playing. On September 30th”*

Gold label: positive; **Predicted:** neutral;

Hypothesis: The phrase “**SO excited**” is a strong positive cue, but because CountVectorizer removes stopwords and lowercases text, “SO” becomes “so”, and punctuation-driven emphasis (“SO” + caps + ellipsis) is lost. The remaining words (“checked”, “events”, “arena”, “playing”) decrease the relative weight of the actual sentiment-bearing term. The model therefore incorrectly treats the tweet as informational/event-related.

3. **Tweet:** *“U2. Dublin. November. Who’s going ? #me”*

Gold label: neutral; **Predicted:** positive;

Hypothesis: The baseline classifier misinterprets the enthusiasm implied by the hashtag **#me** or the structure of the question as positive sentiment. Short tweets with proper nouns and ambiguous excitement markers tend to be challenging for BOW because they require pragmatic interpretation. The model sees “going” and a hashtag that often signals fandom, pushing it toward the positive class – even though the intended sentiment is merely informational (“who’s attending”).

I improved the classifier mainly by making the features smarter and the SVM less overconfident. The baseline model used a simple unigram bag-of-words representation with raw counts and English stopwords removal, trained with a very large SVM regularisation parameter ($C=1000$). That setup still massively overfitted: it achieved almost perfect performance on the training data (macro-F1 ≈ 0.996) but only around 0.52 macro-F1 on the development and test sets. To fix this, I switched from plain counts to a TF-IDF representation, which downweighs extremely common words and highlights terms that are more discriminative for sentiment. I also extended the n-gram range from unigrams to unigrams+bigrams, which lets the model capture short phrases such as “so excited”, “not good”, “hate this”, “love it”, and similar combinations that are crucial in tweets but invisible to a pure unigram model. On top of that, I deliberately kept stopwords so that negation words like “not” and “no” are preserved instead of being stripped out, since these are often the difference between positive and negative polarity. Finally, I reduced the SVM’s C value from 1000 to 1.0 and increased the maximum number of iterations, encouraging a softer margin and better convergence rather than pushing the model to perfectly separate the training data at the cost of generalisation.

The improved classifier clearly outperforms the baseline across all key metrics on both the development and test sets. On the dev set, the baseline model achieved a macro-F1 of about **0.524** and accuracy of **0.541**, while the improved model reached a macro-F1 of about **0.606** and accuracy of **0.632**. On the held-out test set, the baseline reached a macro-F1 of **0.523** and accuracy of **0.539**, whereas the improved model achieved a macro-F1 of **0.611** and accuracy of **0.629**, i.e. roughly +0.09 absolute gains in both macro-F1 and accuracy. All three sentiment classes benefited: on the test set, negative F1 increased from **0.449** to **0.529**, neutral from **0.565** to **0.658**, and positive from **0.554** to **0.645**. While still limited by the inherent ambiguity of short tweets, the improved model therefore provides substantially stronger generalisation and is a clear upgrade over the unigram BOW baseline.

There are still several clear ways to push performance further. First, the feature engineering can be made richer: explicitly modelling emojis, emoticons, elongated words (“soooo”), capitalisation (“SO excited”), and punctuation (“!!!”, “?!”) would let the classifier pick up on intensity and informal sentiment cues that TF-IDF on tokens only partially captures. We could also add a small lexicon-based component (using sentiment dictionaries) or use character-level n-grams to help with misspellings and creative slang. Second, instead of relying purely on sparse bag-of-words, we could incorporate dense word or sentence embeddings, for example by averaging pretrained Twitter embeddings or using a sentence transformer; this would help the model generalise across semantically similar words that never co-occur in exactly the same form in the training data. The biggest potential jump, though, would come from moving to transformer-based models such as BERTweet, RoBERTa, or a sentiment-fine-tuned BERT variant, which can be fine-tuned end-to-end on this dataset or used via zero-/few-shot classification. Those models can capture word order, context, and subtle cues like sarcasm much better than linear TF-IDF+SVM, and on typical Twitter sentiment tasks they usually reach much higher macro-F1 than classical pipelines. Finally, more targeted error analysis – for example, clustering misclassified tweets by pattern

(sarcasm, questions, ambiguous sports/news, multi-sentiment tweets) – would suggest additional, focused tweaks such as special handling of rhetorical questions or domain-specific hashtags.

Assignment Reflection

Completing this assignment required bringing together several different components of text technologies – information retrieval evaluation, token-based statistical analysis, topic modelling, and sentiment classification – into a single integrated workflow. Implementing everything in one file was functional but cumbersome; in practice, it would have been far cleaner to modularise the code into separate scripts, especially given the variety of tasks involved. Nevertheless, working through the implementation in a unified environment helped reinforce how these different techniques relate to one another and how they can be combined in larger text-processing systems.

The most significant learning experience came from the classification component, particularly when attempting to improve the baseline sentiment model. This part of the assignment moved beyond simply reproducing lab code and required making informed decisions about feature engineering, model regularisation, and evaluation. Iteratively testing improvements – such as switching from bag-of-words counts to TF-IDF, adding bigrams, keeping stopwords to capture negation, and lowering the SVM's C value – made clear how sensitive real-world classifiers are to preprocessing choices. Seeing the improved macro-F1 score confirmed not just that these modifications worked, but also why they worked, which was much more instructive than merely applying a pre-packaged model.

A notable challenge throughout the assignment was the need to develop enough background knowledge of the Old Testament, New Testament, and Quran in order to meaningfully interpret the MI/χ^2 rankings and the LDA topics. While the computational outputs were straightforward, explaining them required understanding the thematic and linguistic structures of the texts, and this contextual work took more time than expected. It was, however, rewarding: it made the quantitative results feel grounded rather than abstract, and highlighted how technical tools intersect with domain knowledge.

Overall, the assignment was genuinely insightful and extended my understanding of text technologies far beyond what the labs and lectures alone cover. Moving from simple document scoring to deeper linguistic patterns, then to full classification and model improvement, provided a coherent sense of progression. The work also mirrored real-world NLP pipelines, where evaluation, analysis, and classification interact closely.